Final Exam, Theory of Computation 2023

- Books, notes, communication, calculators, cell phones, computers, etc... are not allowed.
- Your explanations and proofs should be clear enough and in sufficient detail so that they are easy to understand and have no ambiguities.
- You are allowed to refer to material covered in lectures (but not exercises) including theorems without reproving them.
- **Do not touch until the start of the exam.**

Good luck!

Name: ______________________ **N° Sciper:** ______________

Problem 1 / 15 points	Problem 2 / 15 points	Problem 3 / 20 points	Problem 4 / 20 points	Problem 5 / 15 points	Problem 6 / 15 points

Total / 100

1 *(15 pts)* **Quick-fire round.** Consider the following statements.

1. Every regular language is in **P**.
2. If $A \subseteq B \subseteq C$ and both A and C are regular, then B is regular.
3. If A and B are regular, then $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$ is regular.
4. If $A \leq_m \text{HALT}$ and $\overline{A} \leq_m \text{HALT}$, then A is decidable.
5. If A is recognisable, then $\overline{A}$ is recognisable.
6. If $\text{HALT} \leq_p A$, then $A \notin \mathbf{NP}$.
7. If $A \in \mathbf{NP}$ and $B \leq_p A$, then $B \in \mathbf{NP}$.
8. Graph-Isomorphism is **NP**-complete.
9. $\text{SAT} \notin \mathbf{coNP}$.
10. Every CNF formula of size n can be equivalently written as DNF formula of size $O(n^2)$.
11. The language $\{(ab)^n : n \text{ is divisible by } 2023\}$ is regular.
12. The language $\{\langle M \rangle : M \text{ is a TM that halts on all inputs}\}$ is recognisable.
13. The language $\{\langle \varphi \rangle : \varphi \text{ is a CNF such that } \varphi(x) = \varphi(x') \text{ for all inputs } x, x'\}$ is in **NP**.
14. The language $\{\langle p \rangle : p \text{ is an } n\text{-variate polynomial s.t. } p(x) = 0 \text{ for some } x \in \mathbb{Z}^n\}$ is decidable.
15. The language $\{\langle D, 1^n \rangle : D \text{ is a DFA that rejects some } n\text{-bit string } x \in \{0,1\}^n\}$ is in **P**.
(*Compare this to Problems 3 and 5 below!* ☺)

For each box below, write one of the following symbols:

- **T** if the statement is known to be true.
- **F** if the statement is false *or not known to be true*. E.g., both $\mathbf{P} = \mathbf{NP}$ and $\mathbf{P} \neq \mathbf{NP}$ should be marked **F**.
- or leave the box empty.

A correct **T**/**F** answer is worth +1 point, an incorrect answer is worth −1 point, and an empty answer is worth 0 points.

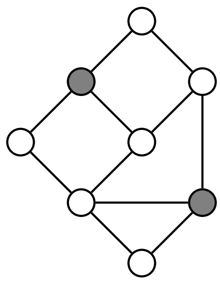
2 *(15 pts)* **Regular languages.** Let $D = (Q, \Sigma, \delta, q_0, F)$ be a DFA. Show that $L(D)$ is infinite if and only if D accepts some string $x \in \Sigma^*$ whose length $|x|$ satisfies $|Q| \leq |x| \leq 2|Q|$.

3 *(20 pts)* **NFAs and NP-completeness.** Show that the following language is **NP**-complete:

$$R_{\text{NFA}} \;=\; \big\{\langle N, 1^n\rangle : N \text{ is an NFA that rejects some } n\text{-bit string } x \in \{0,1\}^n\big\}.$$

(Hint: Reduce from SAT*: given a CNF formula φ over n variables, show how to construct a polynomial-size NFA N such that $\varphi(x) = 1$ iff N rejects x.)*

4 *(20 pts)* **NP-completeness.** Let $G = (V, E)$ be an undirected graph. We say that $S \subseteq V$ is a *blocking set* if it contains at least one node from each cycle in G. In other words, if we remove the nodes S (and all edges adjacent to S) from G, then we are left with a *forest* (graph with no cycles). For example, the highlighted nodes below form a blocking set of size 2.



Show that the following problem is **NP**-complete:

$$\text{Block} \;=\; \{\langle G, k\rangle : G \text{ is a graph that contains a blocking set of of size } k\}.$$

(You may assume the **NP***-completeness of any of the problems discussed in the lectures:* SAT, Independent-Set, Clique, Vertex-Cover, Set-Cover, Subset-Sum*, etc. Make sure to prove that your reduction is correct!)*

5 *(15 pts)* **Undecidability.** Consider the language

$$R_{\mathrm{TM}} = \big\{\langle M, 1^n\rangle : M \text{ is a TM that rejects some } n\text{-bit string } x \in \{0,1\}^n\big\}.$$

Classify R_{TM} as one of (i) decidable, (ii) undecidable but recognisable, (iii) unrecognisable. Justify your answer with a proof.

6 *(15 pts)* **Busy Beaver.** Define the n-th *Busy Beaver* number, denoted $\mathrm{BB}(n)$, as the largest number $k \in \mathbb{N}$ such that there exists a TM $M = (Q, \Sigma, \Gamma, \delta, q_0, q_{\text{halt}})$ such that

(1) M has $|Q| = n + 1$ states (that is, n states in addition to its halting state q_{halt}),
(2) M has a binary input/tape alphabet, $\Sigma = \Gamma = \{0, 1\}$,
(3) M, on the empty input ε, halts with 1^k written on its tape.

In other words, amongst all $(n+1)$-state TMs that halt on the empty input, what is the longest all-1 string that it can output? Note that $\mathrm{BB}(n)$ is always finite, as there are only finitely many distinct TMs satisfying (1)–(3) for any given n.

Show that the function $\mathrm{BB}\colon \mathbb{N} \to \mathbb{N}$ is not computable. That is, show that there is no TM that on input n will always halt with $\mathrm{BB}(n)$ on its tape.

(*Hint: If* BB *were computable, how would you decide the Halting problem?*)